Chapter 1: The Real and Complex Number Systems

Section: Ordered Sets

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Definitions and Setup

Supremum and Infimum

Examples

The Least-Upper-Bound Property

Theorem 1.11 and Proof

Summary

Why This Section Matters

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- We need precise language: *order*, *bounds*, *supremum*
- The least-upper-bound property will distinguish $\mathbb{R}$ from $\mathbb{Q}$

Definitions and Setup

Definition 1.5: Order on a Set

Definition 1.5 — Order

Let S be a set. An order on S is a relation, denoted by $<$, with:

(i) Trichotomy: If $x \in S$ and $y \in S$, then exactly one holds:

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Notation: $y > x$ means $x < y$. $\quad x \leq y$ means $x < y$ or $x = y$.

Definition 1.6: Ordered Set

Definition 1.6 — Ordered Set

An ordered set is a set S in which an order is defined.

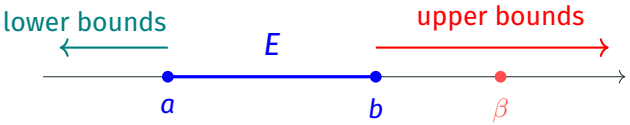
Example: $\mathbb{Q}$ is an ordered set, where $r < s$ means $s - r$ is a positive rational.

Definition 1.7: Upper and Lower Bounds

Definition 1.7 — Bounded Above

Let S be an ordered set and $E \subset S$. If there exists $\beta \in S$ such that $x \leq \beta$ for every $x \in E$, then E is bounded above, and β is an upper bound of E.

Lower bounds are defined symmetrically (with $\geq$ in place of $\leq$).



Supremum and Infimum

Definition 1.8: Least Upper Bound (Supremum)

Definition 1.8 — Supremum

Let S be an ordered set, $E \subset S$, E bounded above. An element $\alpha \in S$ is the least upper bound (or supremum) of E if:

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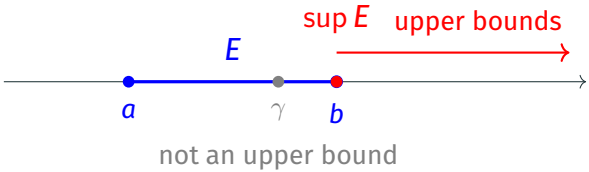
(i) α is an upper bound of E.

(ii) If $\gamma < \alpha$, then γ is *not* an upper bound of E.

We write $\alpha = \sup E$.



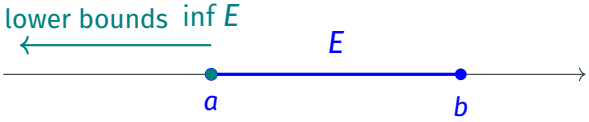
Definition 1.8: Greatest Lower Bound (Infimum)

Definition 1.8 — Infimum

The greatest lower bound (or infimum) is defined symmetrically:
$\alpha = \inf E$ means:

(i) α is a lower bound of E.

(ii) If $\beta > \alpha$, then β is *not* a lower bound of E.



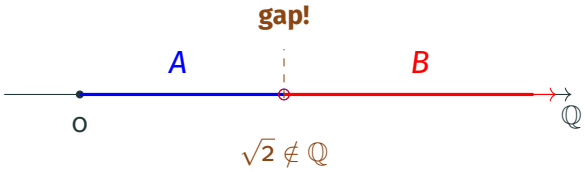
Examples

Example 1.9(a): No Supremum in $\mathbb{Q}$

Example 1.9(a)

Recall the sets from Example 1.1:

$$A = \{p \in \mathbb{Q} : p > 0,\ p^2 < 2\}$$
$$B = \{p \in \mathbb{Q} : p > 0,\ p^2 > 2\}$$



Example 1.9(a): No Supremum in $\mathbb{Q}$

Example 1.9(a) — Conclusion

- The upper bounds of A in $\mathbb{Q}$ are exactly the members of B.
- B has no smallest member $\implies$ A has no supremum in $\mathbb{Q}$
- The lower bounds of B are A and all $r \leq 0$. Since A has no largest member, B has no infimum in $\mathbb{Q}$

Example 1.9(b): Supremum May or May Not Belong to E

Example 1.9(b)

If $\alpha = \sup E$ exists, α may or may not be a member of E.

Let $E_1 = \{r \in \mathbb{Q} : r < 0\}$ and $E_2 = \{r \in \mathbb{Q} : r \leq 0\}$.

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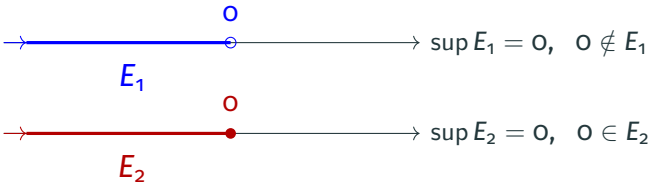
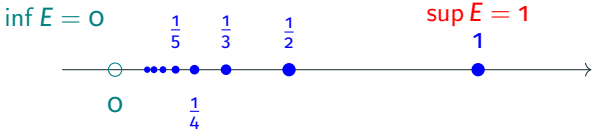


Both have $\sup = 0$, but membership differs.

Example 1.9(c): The Set $\{1/n\}$

Example 1.9(c)

Let $E = \{1/n : n = 1, 2, 3, \ldots\} = \{1,\ 1/2,\ 1/3,\ 1/4,\ \ldots\}$



$\sup E = 1 \in E, \quad \inf E = 0 \notin E.$

The Least-Upper-Bound Property

Definition 1.10: Least-Upper-Bound Property

Definition 1.10 — Least-Upper-Bound Property

An ordered set S has the least-upper-bound property if:

Whenever $E \subset S$, $E \neq \varnothing$, and E is bounded above, then $\sup E$ exists in S.

Example 1.9(a) shows: $\mathbb{Q}$ does *not* have the least-upper-bound property.

Theorem 1.11 and Proof

Theorem 1.11: LUB Property Implies GLB Property

Theorem 1.11

Suppose S is an ordered set with the least-upper-bound property, $B \subset S$, $B \neq \varnothing$, and B is bounded below. Let L be the set of all lower bounds of B.

Then $\alpha = \sup L$ exists in S, and $\alpha = \inf B$.

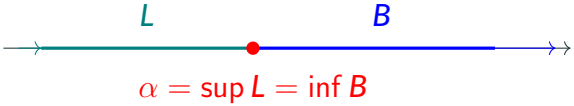
In particular, $\inf B$ exists in S.

Proof Strategy: Theorem 1.11

Goal: Show $\sup L = \inf B$, where $L = \{\text{lower bounds of } B\}$.

Approach:

1. Show L is nonempty and bounded above $\implies \sup L$ exists.
2. Show $\alpha = \sup L$ is a lower bound of B (i.e., $\alpha \in L$).
3. Show nothing larger than α is a lower bound of B.
4. Conclude $\alpha = \inf B$.



Proof of Theorem 1.11 — Step 1: $\sup L$ Exists

Step 1: L is nonempty and bounded above

- B is bounded below $\implies L \neq \varnothing$.
- Every $x \in B$ is an upper bound of L (since $y \leq x$ for all $y \in L$, $x \in B$).
- L is bounded above $\implies \sup L$ exists in S (by the LUB property).

Call this supremum $\alpha = \sup L$.

Proof of Theorem 1.11 — Step 2: α Is a Lower Bound of B

Step 2: $\alpha \in L$

Take any $x \in B$. Since x is an upper bound of L and $\alpha = \sup L$:

$$\alpha \leq x \quad \text{for every } x \in B.$$

Therefore α is a lower bound of B, i.e., $\alpha \in L$.

Proof of Theorem 1.11 — Step 3: α Is the Greatest Lower Bound

Step 3: No $\beta > \alpha$ is a lower bound of B

If $\alpha < \beta$, then $\beta \notin L$.

Why? α is an upper bound of L, so every $y \in L$ satisfies $y \leq \alpha < \beta$. If β were in L, we'd need $\beta \leq \alpha$ — contradiction.

Conclusion:

- α is a lower bound of B (Step 2)
- No $\beta > \alpha$ is a lower bound of B (Step 3)

$\boxed{\alpha = \inf B}$

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- An order on a set requires trichotomy and transitivity (Def. 1.5–1.6)

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- $\mathbb{Q}$ lacks the least-upper-bound property (Ex. 1.9, Def. 1.10)

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- The supremum is the least upper bound; the infimum is the greatest lower bound (Def. 1.7–1.8)
- $\mathbb{Q}$ lacks the least-upper-bound property (Ex. 1.9, Def. 1.10)
- LUB property $\implies$ GLB property (Thm. 1.11)

Coming up next: Fields — the algebraic structure of $\mathbb{Q}$ and $\mathbb{R}$.